

Improving Preparedness of Communities for Evacuations using ZEVs

Development of ZEV Evacuation Readiness Score

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Background

- What comes to mind when we tend to compare two things?
- Rating/Ranking is the go-to method to rank two products of same kind:



Iphone 15 pro max

Rating: ★★★★★

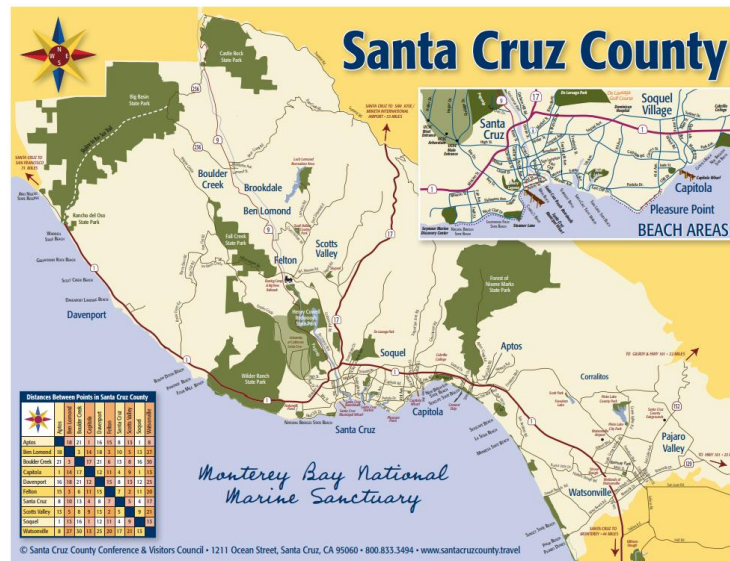


Samsung S24 ultra

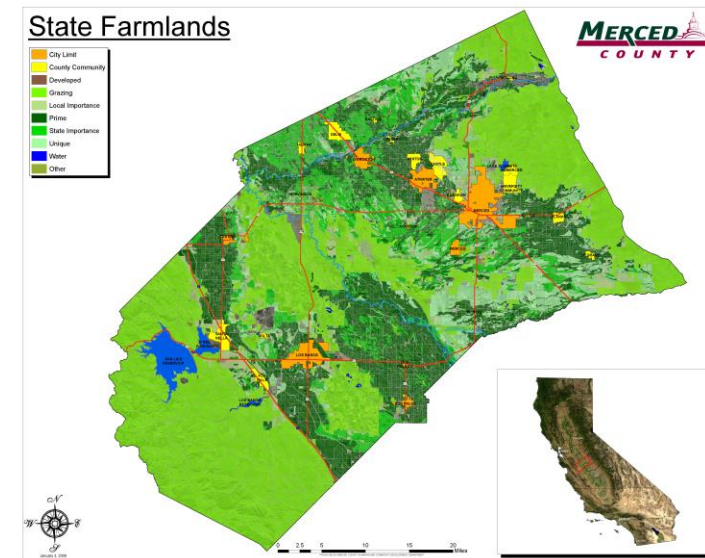
Rating: ★★★★★

Introduction to ZEV Evacuation Score

- ZEV evacuation readiness score provides a basis to categorize each county based on its preparedness to successfully carry out a zero-emission vehicle evacuation.



ZEV Score: ★★★★★

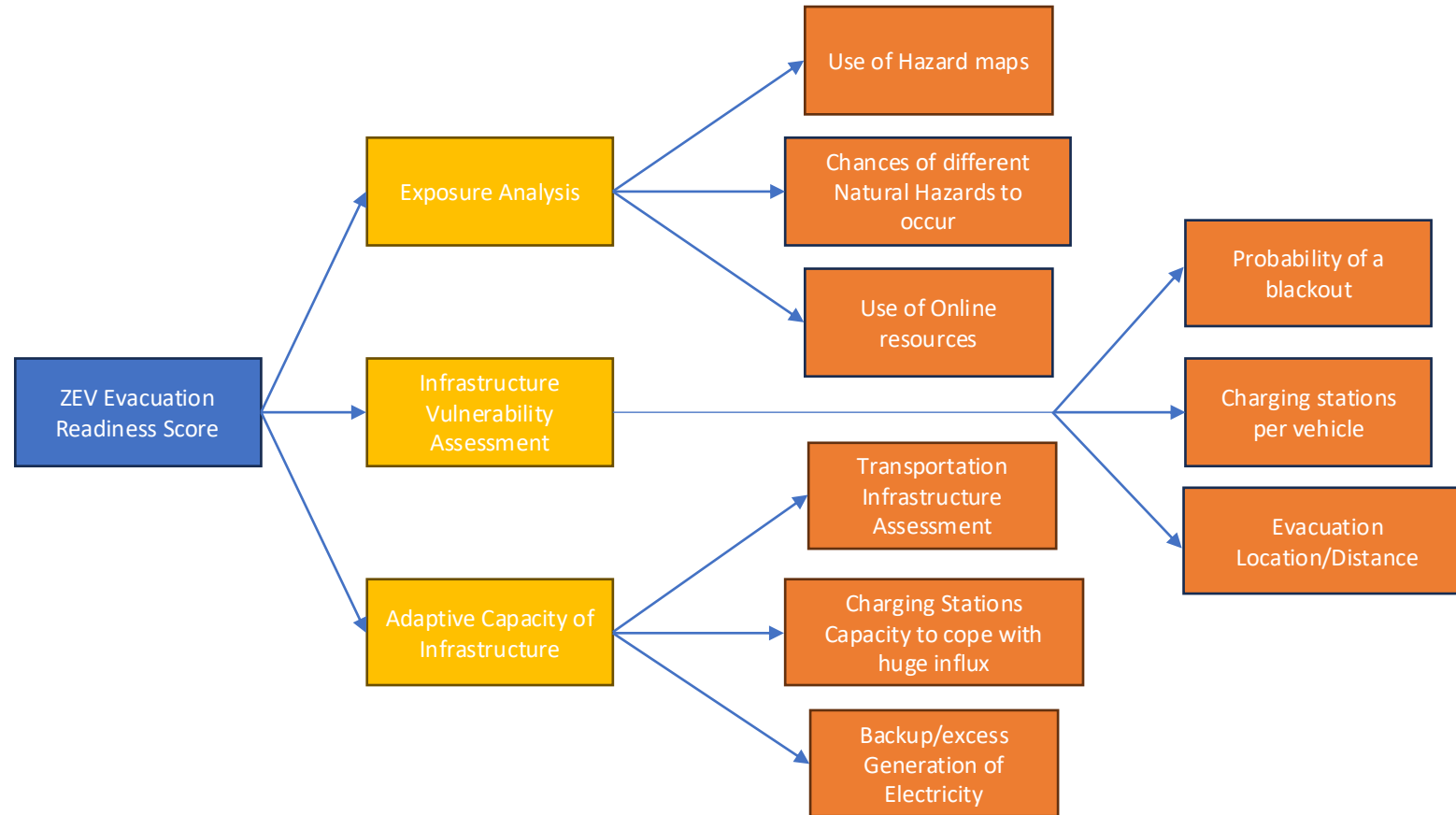


ZEV Score: ★★★★★

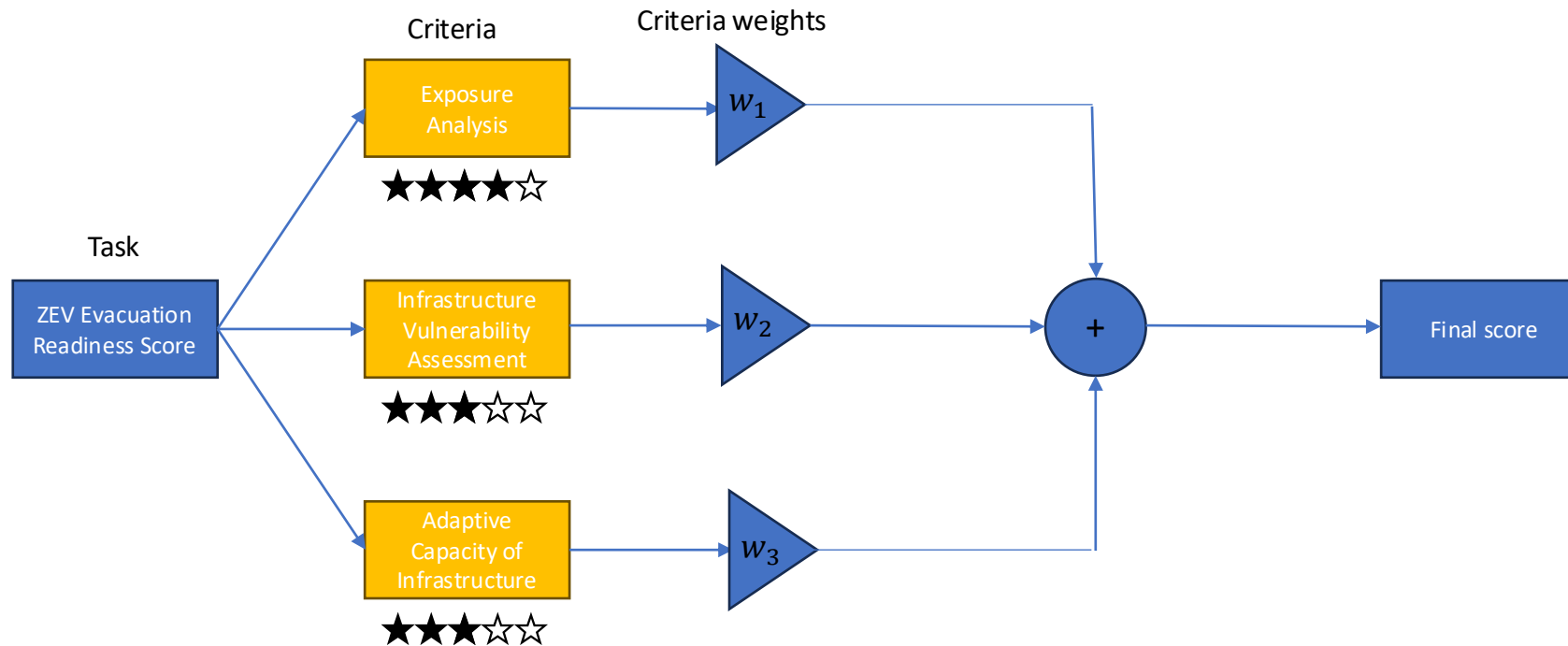
Importance of ZEV Evacuation Score

- This score can help us identify the indicators that need more research and development in a certain county.
- It can also aid public organizations in directing technical and financial support to communities that need more help.
- As this score depends upon different indicators. Certain score can be broken down into its specific indicator indicating which requires more development and funding.

Indicators used in ZEV Evacuation Score



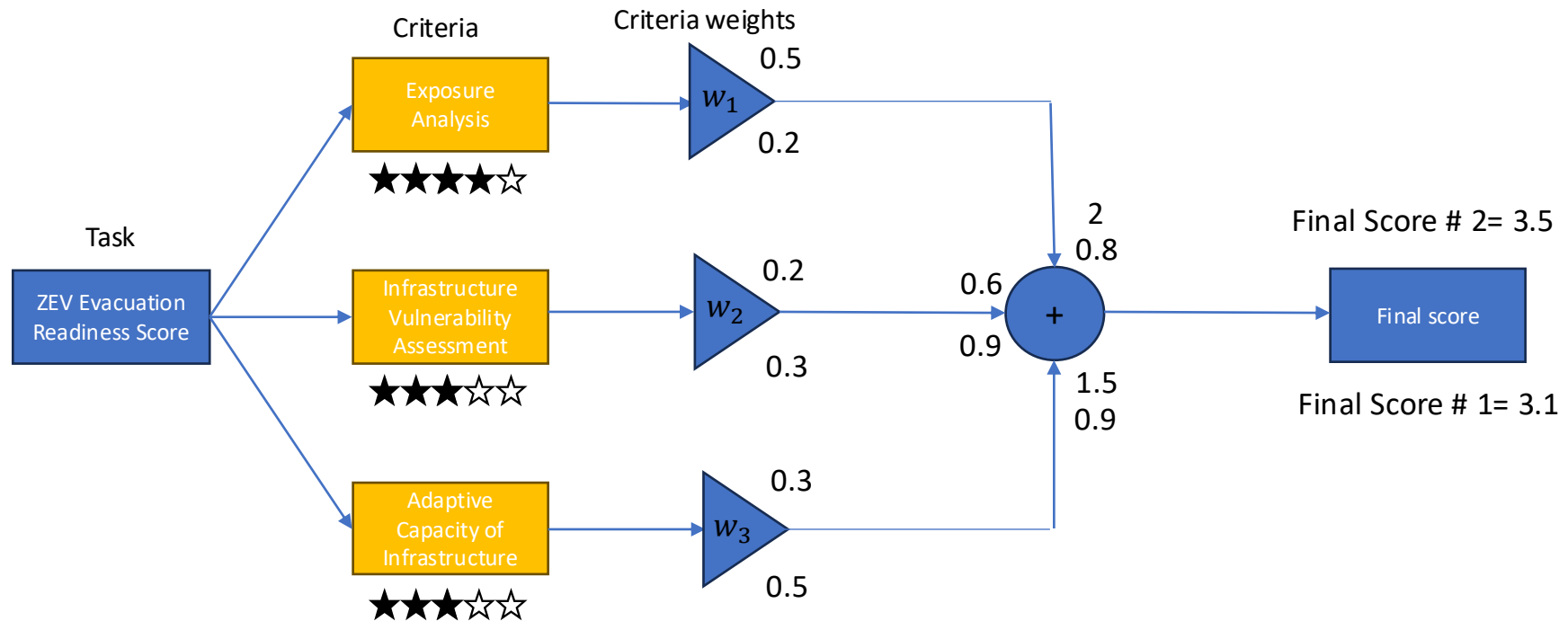
Development of ZEV Score From Indicators



- The weights in the above flowchart depends on the survey result from the community and experts (YOU!).
- How would it affect the score? Let's see!

Importance of Criteria Weights

- Let's calculate ZEV Score for different weights:

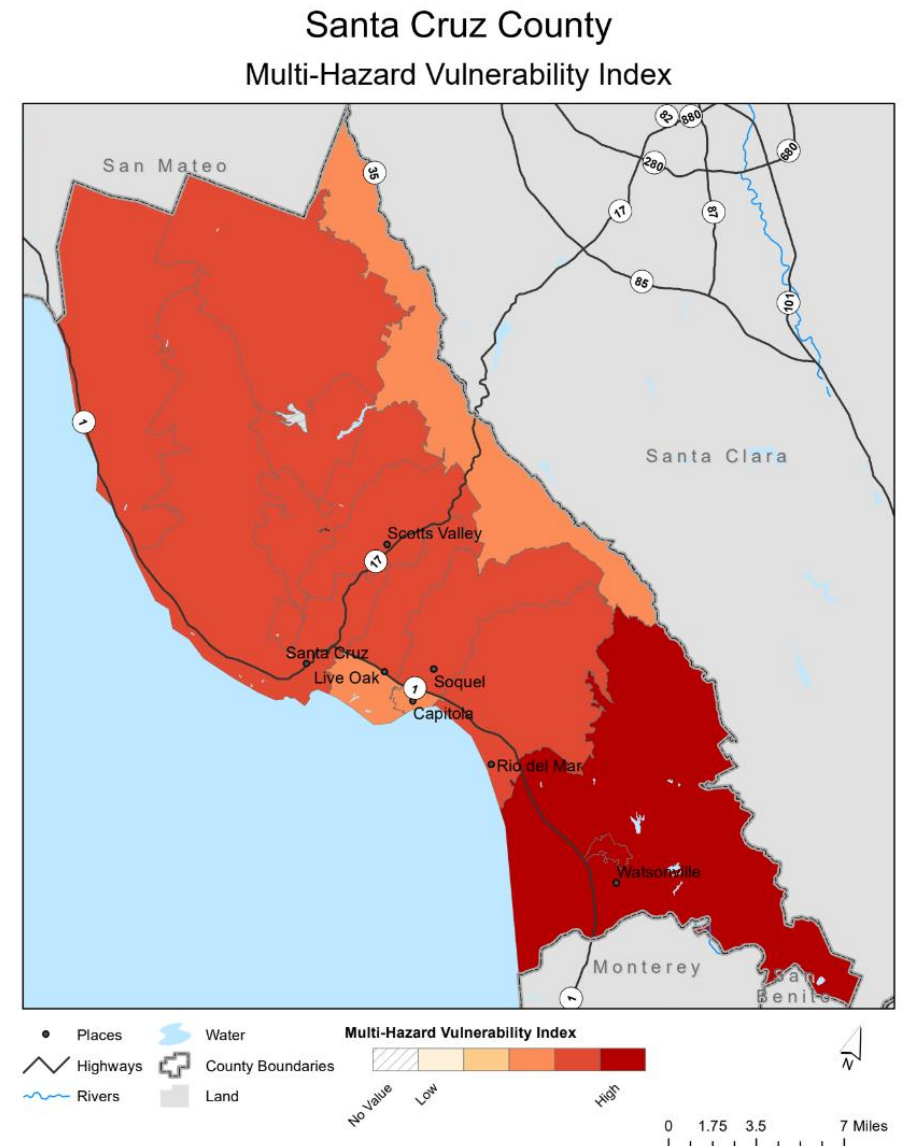


- Let's see what different criteria are in detail.

Exposure Analysis (EI)

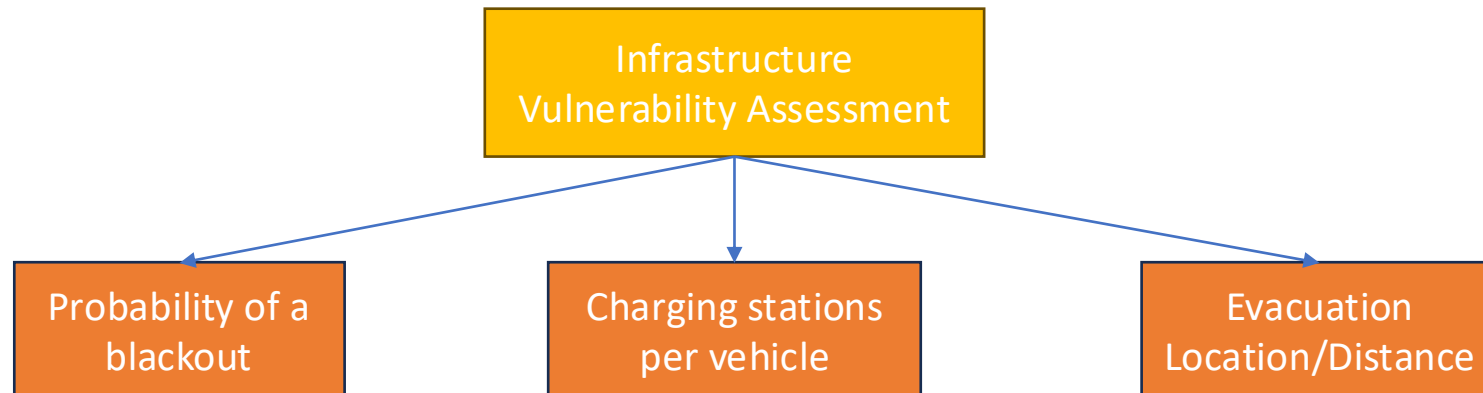
- This criterion deals with the estimation of the level to which the area/county is prone to certain hazards.
- These hazards can include wildfires, floods, earthquakes, or an overall estimation of all kinds of hazards.
- In our study, we are giving a score to each county based on maps developed for hazard analysis and estimation methods.

Exposure Index: 0.8872975397361342



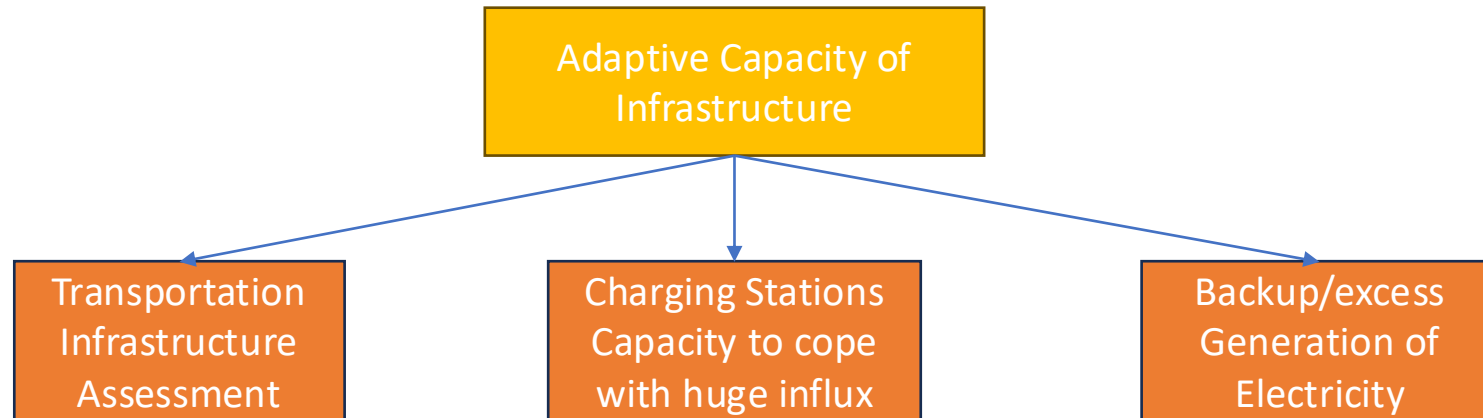
Infrastructure Vulnerability Assessment (IVS)

- This criterion deals with the assessment of already available infrastructure.
- This infrastructure include Electric chargers, roads, electricity transmission network etc.
- We are assessing the ability of area/county to deal with any natural hazards with already built/developed infrastructure

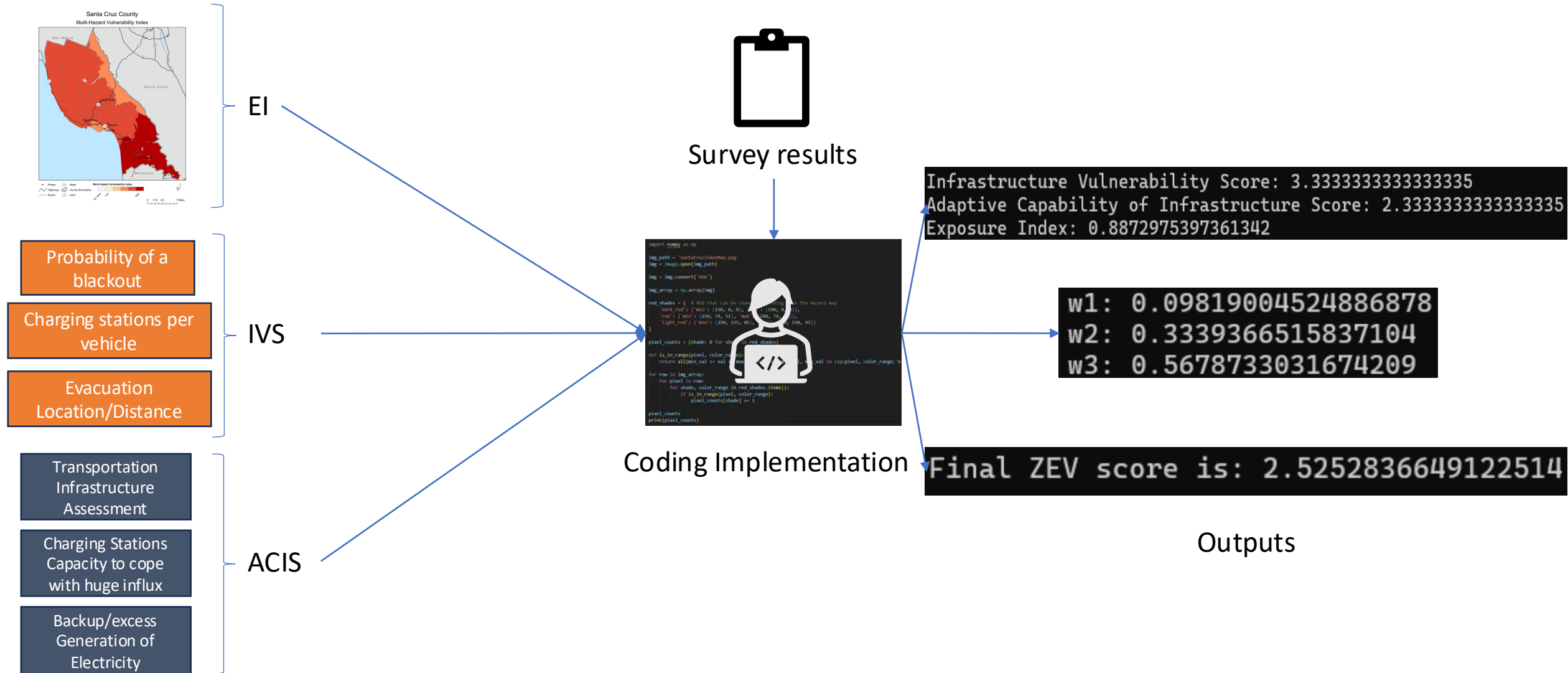


Adaptive Capacity of Infrastructure (ACIS)

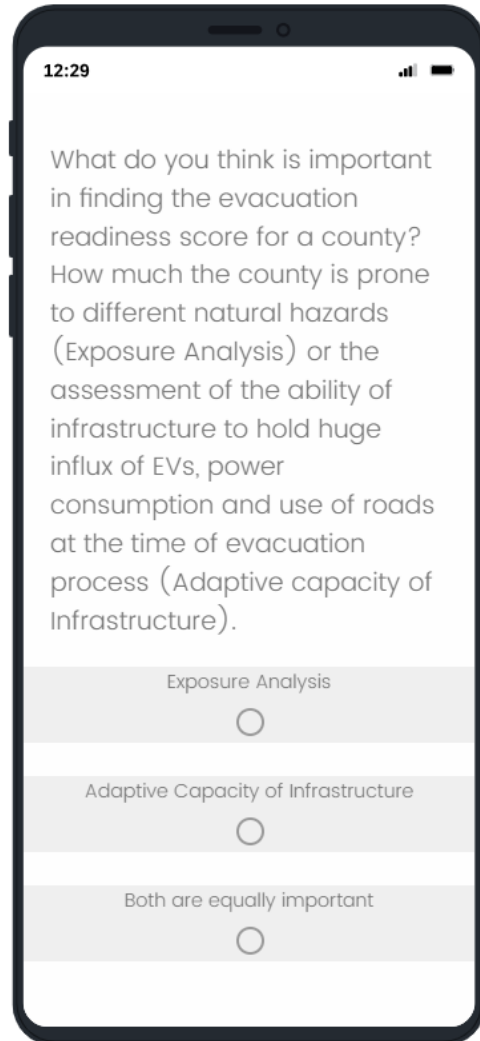
- This criterion deals with the assessment of how adaptive the already available infrastructure is in case of any natural hazard.
- In this criterion, we estimate the ability of electric chargers, power generation units, power transmission systems and roads to take up load and stress in case of evacuation.



Development of a Coding Algorithm



Survey's Overview



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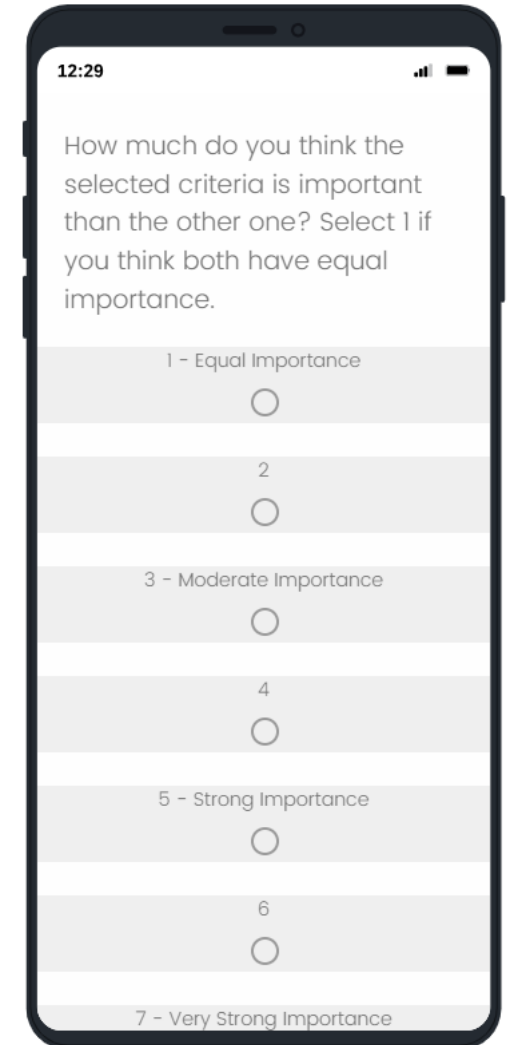
What do you think is important in finding the evacuation readiness score for a county? How much the county is prone to different natural hazards (Exposure Analysis) or the assessment of the ability of infrastructure to hold huge influx of EVs, power consumption and use of roads at the time of evacuation process (Adaptive capacity of Infrastructure).

Exposure Analysis ☐

Adaptive Capacity of Infrastructure ☐

Both are equally important ☐

- There will be 6 questions in total (Estimated time to complete survey: 8-10 mins)
- 1st question deals with the relative importance of different criteria.
- 2nd question delves into deeper comparison and asks for a rating signifying the importance of selected criterion as compared to the other one.



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How much do you think the selected criteria is important than the other one? Select 1 if you think both have equal importance.

1 - Equal Importance ☐

2 ☐

3 - Moderate Importance ☐

4 ☐

5 - Strong Importance ☐

6 ☐

7 - Very Strong Importance ☐

Survey's QR Code



Thank you!

